Math 515 Problem Set 4

Prabash Premarathna

February 2026

3.

(a) The characteristic polynomial is

$$\rho(\lambda) = \lambda^3 - \frac{18}{11}\lambda^2 + \frac{9}{11}\lambda - \frac{2}{11}.$$

Since $\lambda = 1$ is a root,

$$\rho(\lambda) = (\lambda - 1)\left(\lambda^2 - \frac{7}{11}\lambda + \frac{2}{11}\right).$$

The remaining roots are

$$\lambda = \frac{7 \pm i\sqrt{39}}{22},$$

and

$$|\lambda| = \sqrt{\frac{2}{11}} < 1.$$

Thus all roots satisfy $|\lambda| \leq 1$, and the root $\lambda = 1$ is simple. Hence the method is zero-stable (strongly stable).

(b) The local truncation error is

$$\tau_{j+1} = y(t_{j+1}) - \frac{18}{11}y(t_j) + \frac{9}{11}y(t_{j-1}) - \frac{2}{11}y(t_{j-2}) - \frac{6}{11}h\,y'(t_{j+1}).$$

Expand about t_{j+1}:

$$y(t_j) = y - hy' + \frac{h^2}{2}y'' - \frac{h^3}{6}y''' + O(h^4),$$

$$y(t_{j-1}) = y - 2hy' + 2h^2y'' - \frac{4}{3}h^3y''' + O(h^4),$$

$$y(t_{j-2}) = y - 3hy' + \frac{9}{2}h^2y'' - \frac{9}{2}h^3y''' + O(h^4).$$

Substituting and collecting terms:

Constant terms:

$$1 - \frac{18}{11} + \frac{9}{11} - \frac{2}{11} = 0$$

y' terms:

$$\frac{18}{11}h - \frac{18}{11}h + \frac{6}{11}h - \frac{6}{11}h = 0$$

y'' terms:

$$-\frac{9}{11}h^2 + \frac{18}{11}h^2 - \frac{9}{11}h^2 = 0$$

Thus all terms up to order h^2 cancel, and the first nonzero term is of order h^3. Hence

$$\tau_{j+1} = O(h^3).$$

Since $\tau_{j+1} \to 0$ as $h \to 0$, the method is consistent.

(c) Since the method is both zero-stable and consistent, it follows that the method is convergent.